# An Empirical Approach to the Distributional Impact of Monetary Policy in Highly Unequal Emerging Economies: The Case of Colombia

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Following the Bayesian approach, SVAR models are estimated for the case of Colombia, a small open emerging economy with high inequality. The sign restrictions methodology for the identification of structural shocks defines the impulse response function (IRF) analysis that seeks to determine the impact of monetary policy (MP) shocks on inequality. Eight different measures of inequality are separately included to estimate different models and their corresponding IRF. The median IRF of all inequality measures show that, after a contractionary MP shock, inequality tends to increase. However, once the credible sets are taking into account, the evidence on the true direction of the response significantly weakens. Although it might be the case that the effect of contractionary MP shocks on inequality is negative, a Forecast Error Variance Decomposition (FEVD) analysis show that the relative importance of MP shocks to explain inequality in Colombia is quite small.

# 1. Introduction

In the developed world, the aftermath of the financial crisis of 2008 set the ground for a massive amount of criticism to the field of macroeconomics and its inability to provide governments and central banks with the appropriate tools to predict and anticipate the development of the crisis. The post-crisis period was also the perfect time to witness the rebirth of a massive concern on inequality, not only from the long-run perspective but also from the point of view of its relation with macroeconomic policies. The structural differences between developed and emerging economies have to be part of that concern in order to understand such a relationship in the context of a more connected world (Prasad 2014).

Traditionally, through the use of tax and transfers policies, an active role has been given to fiscal policy to have an impact on the distribution of income. On the contrary, monetary policy have had only an indirect role through the control of the inflation rate. After the crisis, such traditional point of view has radically changed as part of a strong and critical assessment of the role played by the monetary authorities of the world in failing to prevent the strong consequences of the economic contraction.

Around these concerns, an important amount of both theoretical and empirical literature has been produced trying to get a deeper understanding of the possible impact of monetary policy on inequality, and there has been some consensus on identifying at least five different channels through which changes in the monetary policy instrument could affect the distribution of income.[1] The first channel has been called the *income composition channel*: since there is heterogeneity in the sources of income across households in the economy, and since the share of financial income of high-income households is higher than that of low-income households, an increase in the intervention interest rate of the central bank (contractionary monetary policy) might decrease inequality by reducing the price of financial asset prices.

A second related channel is the so-called *financial segmentation channel*: high-income households are prone to have more and frequent connections with the financial system, making them more vulnerable to the risks associated with a contractionary monetary policy shock, which would, therefore, reduce inequality.

[1]These channels have been summarised by Coibion et al. (2017); Davtyan (2017); Dolado, Motyovszki, and Pappa (2018) and Furceri, Loungani, and Zdzienicka (2018).

A third channel, through which a contractionary stance of monetary policy could also have a positive impact on the distribution of wealth and income is the *portfolio channel*: since it is likely that low-income households hold a greater share of its income/wealth in the form of cash, any deflationary action from the central bank, would reduce not only income inequality but consumption inequality as well.

On the contrary, an increase in the intervention interest rate could increase inequality through the two remaining channels. First, given the fact that low-income households tend to be net borrowers, a surprise increment of the intervention rate and the consequent unexpected decrease in the inflation rate, would generate a wealth transfer from net borrowers (low-income households) to net savers (high-income households) through the *savings redistribution channel*. Finally, since the share of labour income on total earnings is relatively greater for low-income households, and that it might be the case that unemployment falls unevenly hard on this group of the population, any countercyclical contractionary monetary policy shock would increase inequality through the *earnings heterogeneity channel*.

It is hard to unveil the workings of each one of these channels though, and the effects of monetary policy on inequality may not have a unique direction since the inner characteristics of the distribution of income and the nature of the monetary policy actions can be quite different among countries. In the particular case of emerging developing economies, it has been argued that the distributional consequences of monetary policy might be of first order of importance, given the underdevelopment of their financial markets (Prasad and Zhang 2015). Likewise, it has been found that negative impacts on inequality from contractionary monetary policy are more likely to be found in economies with low redistribution and high initial levels of income inequality (Furceri, Loungani, and Zdzienicka 2018).

Both from the theoretical and empirical approximations, research on this subject has been strongly biased towards the situation of developed economies. Up until the last of my knowledge, the only attempt to establish the impact of monetary policy shocks on inequality in an emerging economy has been Villarreal (2016) for the case of Mexico. Among the emerging economies, however, Colombia is perhaps one of the most unequal ones. According to official data from the National Statistics Department (DANE), in the last 15 years the Gini coefficient has not been lower than 0.5, and has shown a really slow decreasing trend.[2] Within

[2] Official figures show that the value of the Gini coefficient was 0.572 in 2002. In 2017, 15 years later, the value had decreased only to 0.508.

the roots of this high and persistent income inequality, the increasing returns to high education, the strong barriers to the access to this kind of education, the high burden of unemployment on the less educated segment of the population, and a strong shift of the labour force to the informal economy, have been identified as the most important factors (Attanasio, Goldberg, and Pavcnik 2004; Joumard and Londoño-Vélez 2013; Aristizábal-Ramírez, Canavire-Bacarreza, and Jetter 2015). Interestingly as well is the fact that from 2001 onwards the Colombian economy adopted an inflation targeting regime to conduct monetary policy, using its intervention interest rate in the short-term interbank market for securities as the main policy instrument. Since then, the posture of the Colombian central bank has been characterised for its countercyclical stance.

The context makes Colombia a suitable subject to test for the impact of monetary policy (MP) shocks on inequality in a highly unequal emerging economy and creates an interesting opportunity to make a contribution to the field. I present an empirical approach based on a Bayesian Structural VAR (SVAR) model for a small open economy. The model includes as main macroeconomic endogenous variables the (log of) real GDP; the intervention rate of the central bank; the rate of inflation of the Consumer Price Index (CPI), and the (log of) the nominal exchange rate. Since from a macroeconomic point of view, the burden of unemployment may be behind the dynamics of inequality, the unemployment rate is also included as an endogenous variable in the empirical small open economy model.

Along the six macroeconomic variables listed above, an inequality variable is also included in the set of endogenous variables. In the baseline model, the identification of the structural monetary policy (MP) shock is based on the less restrictive and popular sign restrictions approach (Uhlig 2005), which, from an accepted theoretical point of view, helps to give shape to the Impulse Response Functions (IRF) of some of the endogenous variables. From the sign restrictions imposed on the IRF, it is expected that a contractionary MP shock −a one time one standard deviation unexpected increase in the intervention policy rate −has a negative impact on real GDP and inflation for at least one year, and an immediate appreciation effect on the nominal exchange rate. Both the unemployment rate and the measure of inequality are let free to respond to the shock.

Eight different measures of inequality are separately included to estimate different models and the corresponding IRF. The purpose is to determine if monetary policy has a differentiated effect on possibly different aspects of income inequal-

ity. Based on two different measurements of individual income, namely the total income per capita of the household, and the labour income, the first six measures of inequality computed are the Gini coefficients of the total income per capita of the household and the labour income; the ratios of the 90th percentiles and the 10th percentiles of total income per capita and labour income, and the ratios of the 75th percentile and the 25th percentiles of total income per capita and labour income. Since inequality can be founded on the higher premium of educated workers, and the significant shift of a segment of the labor force to the informal sectors, the two other measures of inequality are the wage premium, computed as the ratio between the wages of high-skilled and low-skilled workers, and the formality premium, measured as the ratio between the wages of workers of the formal and the informal sectors.

All variables are available on a quarterly basis, and the values of the real GDP and the unemployment rate are seasonally adjusted. The data sample covers the period from the first quarter of 2008 to the fourth quarter of 2017. This period of time was determined by the availability of reliable time series for the measures of inequality. Each one of these measures was computed using survey data from the Colombian *Great Integrated Household Survey*, which was applied by DANE from the second part of 2006 onwards, after a significative methodological revision took place. However, time-comparable data in terms of individual income is only available starting from the first quarter of 2008. The Bayesian approach to the estimation of the SVAR models was adopted to help overcome the possible small-sample issues, and because the sign restrictions identification approach is usually applied under this methodological framework.

The median IRF of all inequality measures of the baseline model, show that after a contractionary MP shock inequality tends to increase but, once the credible sets are taking into account, the evidence on the true direction of the response significantly weakens, making the response more likely to be positive two years after the shock. Although it might be the case that the effect of contractionary MP shocks on inequality is negative, a Forecast Error Variance Decomposition (FEVD) analysis show that the relative importance of MP shocks to explain inequality in Colombia is quite small.

Two different robustness checks were performed to determine the strength of the results obtained with the baseline model. The first one tested the results changing the number of quarters ahead of the shock for which the restrictions

imposed on the response of real GDP and inflation to the MP shock were kept. For the models with measures of inequality based on total income per capita, the results did not change significatively. Nevertheless, for the case of the measures based on labour income, the median response shape was inverted in the first four quarters following the shock.

Since the effects of the MP shock are somehow defined by the sign restrictions imposed, the second robustness check was to change the methodology of identification of the structural shocks. The triangular factorisation approach (Hamilton 1994) was adopted, which imposes zero restrictions only in the immediate impacts of the endogenous variables of the SVAR model to the structural shocks. The theoretical implications of the zero restrictions depend on the order of the variables inside the vector of endogenous variables, so the order selected was one with close theoretical implications to the sign restrictions imposed in the baseline model. The triangular factorisation makes a strong case for the support for a negative effect of the contractionary MP shock on inequality. However, this last result may be highly related to the significative and negative response of the unemployment rate under this identification approach, and which was not present with the sign restrictions. The evidence presented is not strongly conclusive about the effects of the monetary policy shocks on inequality in Colombia.

The document is divided in six parts, including this introduction. The second part reviews the related existing literature on the impact of monetary policy on inequality. The third one describes carefully the SVAR model, the sign restrictions approach to the identification of structural shocks, and the Bayesian method, whilst the fourth part describes the results of the IRF analysis of the baseline model. The fifth section carries out the robustness checks and explains briefly the triangular factorisation identification methodology. The final section summarises the main conclusions.

## 2. Related literature

It would be possible to classify the literature on the distributional effects of monetary policy in three different but closely related lines of research. For the first of these lines, the general idea is to determine how the dynamics of the main targets of monetary policy –e.g. the rate of inflation and the GDP gap– affect the distribution of income and wealth. In this line of work Romer and Romer (1999),

for the case of the United States (US) and several countries of the OECD in the post-war period, find that surprise inflation episodes, associated with lower unemployment and an increase in the GDP gap, had positive short-run effects on the distribution of wealth and the reduction of poverty. However, based on the theoretical idea that monetary policy has effects only in the short-run, they claim that such positive impacts tend to disappear. By means of a regression analysis relating poverty and distribution indicators to the volatility of inflation and economic growth, Romer and Romer conclude that long-run positive effects on the distribution of wealth and poverty reduction can only be accomplished by sound and systematic monetary policy.

Such an indirect reading on the possible effects of monetary policy on inequality, can also be extracted from the work of Doepke and Schneider (2006). The authors document nominal positions of assets and liabilities for different groups of households in the United States along the post-war period. Their main objective was not to directly relate monetary policy to the distribution of income and wealth, but to determine if moderate inflation episodes could have had important redistributional effects by its impact on the nominal valuation of such financial positions. By performing a counterfactual exercise in which inflation is 5% higher than expected in a period of 10 years –a moderate inflation episode similar to that of the 1970s in the US–they found that rich and old households (net savers) might have suffered loses between 6% and 15% of GDP, whilst middle-class young households (net borrowers) might have had all the gains. Although it is not possible to associate this moderate inflation episode to a particular monetary policy stance, Doepke and Schneider work is the main source to understand the *savings redistribution channel* of monetary policy.

The focus of the second identified line of research is to provide direct evidence regarding the relationship between monetary policy shocks, defined as the unexpected component of the changes in the monetary policy instrument, and indicators of inequality by means of empirical statistical models of time series analysis. For instance, by using the local projection method suggested by Jordà (2005), Coibion et al. (2017) estimate Impulse Response Functions (IRF) for several inequality indicators of the US, including the Gini index for labour and total income and consumption expenditures, to a set of estimated monetary policy shocks[3] based on a quarterly data set covering the period 1969-2008. The IRF anal-

[3]The monetary policy shocks were estimated following the methodology in Romer and Romer

ysis shows that contractionary monetary policy shocks increased all measures of inequality, especially those related to total income and consumption expenditures. Although they recognise that the point estimates of the effect might be rather imprecise, forecast error variance decompositions allow them to state that monetary policy shocks account for at least 10% of the variance of income inequality, and around 20% of the variance of consumption inequality.

Davtyan (2017) argues that the result of increasing inequality as a consequence of contractionary monetary policy shocks relies heavily on the lack of data regarding the top 1% of the US income distribution in the data set used by Coibion et al.. Using the Gini index and the 90th to 10th percentile ratio computed by the OECD for a measure of the household's disposable income on a yearly basis, which contains information of the top 1% population of the distribution of the US income, Davtyan estimates a Vector Error Correction (VEC) model for quarterly data for the period 1983-2012. By imposing zero long-run and short-run restrictions to identify the structural monetary policy shock, the IRF analysis finds that, contrary to the findings of Coibion et al. (2017), contractionary monetary policy shocks in the US reduced overall income inequality.[4]

For the case of the United Kingdom, Mumtaz and Theophilopoulou (2017) estimated a SVAR model using quarterly data between 1969-2012, in order to determine if there were any impacts of the monetary policy shocks on earnings, income and consumption inequality. The sign restrictions approach of Uhlig (2005) was used to identify the structural monetary policy shocks and reliable IRF based on the belief that a contractionary shock should lead to an increase in the interest rate, an appreciation of the nominal exchange rate, and decreases in both the real GDP and the CPI inflation rate. The IRF analysis isolated the possible differential effects of monetary policy shocks before and after the use of Unconventional Monetary Policy (UMP), and found that contractionary monetary policy shocks increased the wage, income and non-durable and total consumption Gini coefficients. In a two-year horizon forecast period, the contribution of the shock was approximately 10% for the income Gini index and 20% for the non-durable consumption Gini

(2004).

[4]Two things should be taken into account when contrasting Coibion et al. and Davtyan results. First, in the latter case, the sample includes the period of Unconventional Monetary Policy (UMP) measures of the Federal Reserve after the financial crisis of 2008 while estimations of the former are based on a sample previous to this period. Second, in order to obtain quarterly data on inequality, Davtyan (2017) interpolates both yearly measures of inequality, a procedure that might cast some doubt on the reliability of the data on the response variable.

index.

Estimating a SVAR model as well, Saiki and Frost (2014) reviewed the impact of UMP on inequality for the case of Japan using a quarterly data set for the period 2008-2013. Identification of the monetary policy shocks was done by imposing zero restrictions on the structural response matrices, which implied a lagged impact of UMP shocks to GDP and prices; an immediate effect of GDP and prices shocks on UMP shocks; and an immediate impact of UMP shocks on stock prices and inequality measures. IRF analysis found that UMP shocks increased inequality, a result that Saiki and Frost informally rationalised by stating that during this period, japanese high-income households in the top 20% of the distribution of income held massive holdings of equities that were overvalued by the Bank of Japan's UMP measures. Inequality was then affected through the *income composition channel*.

From a more general point of view, Furceri, Loungani, and Zdzienicka (2018) estimated IRF for the Gini coefficient, the top 10%, 5% and 1% income shares, and the share of wage income using a sample from a panel of 32 advanced and emerging economies in the period 1990-2013. Estimation was based on the local projection method of Jordà (2005) and the identification of monetary policy shocks followed an approach close to the one proposed by Romer and Romer (2004) for the case of the US. The IRF analysis based on the whole panel sample indicates that contractionary monetary policy shocks increased inequality, and the result is robust across the different inequality measures used. By creating two subsamples based on the sign of the difference between the value of the Gini coefficient and the average value across countries, IRF functions show that the increase of inequality generated by contractionary monetary policy shocks, is highly significative for relatively unequal countries, and it is not statistically significant for countries with relatively high redistribution.

Perhaps the only attempt so far that focuses on determine the impact of monetary policy on inequality in an emerging market economy is Villarreal (2016). The process of estimation of the impact is carried over following two steps. First, a Bayesian estimation of an open-economy DSGE model for the Mexican economy is provided to identify monetary policy shocks; second, such estimated shocks are then used to estimate local projections of IRF (Jordà 2005) for the Gini of labour income. Quarterly data used for both stages of the estimation covered the period 1995-2012, which coincided with the period of adoption and implementation of an inflation targeting regime for monetary policy in Mexico. The IRF analysis

provided by Villarreal established that an unexpected increase in the intervention monetary policy rate (contractionary monetary policy shock) reduced the Gini of labour-income. A robustness check was performed estimating an SVAR model for which monetary policy shocks were identified imposing log-run and short-run sign restricitions (Uhlig 2005). The estimated IRF confirmed the main result.

In the third and final identified line of research, the methodological approach goes beyond statistical data analysis and relies on estimated theoretical heterogeneous agents DSGE models to try to either formally rationalise the empirical results of time series models or directly build IRF and determine the impact of monetary policy shocks on inequality. Starting from a pure theoretical approach, Gornemann, Kuester, and Nakajima (2016) build a New-Keynesian (NK) sticky-price DSGE model with search and matching frictions in the labour market, incomplete financial markets, and rich heterogeneity based on differences in employment status, productivity and patience. They calibrate the model to quarterly data of the US economy in the period 1981-2008 in order to replicate the main characteristics of the US income and wealth distribution. IRF from the numerical solution of the model shows that under such a framework, a contractionary monetary policy shock increase the inequality of earnings, income, wealth and consumption. The rationalisation of the result inside the model is determined by the capacity of high-income households to insure themselves against unemployment risks by holding a diversified portfolio.

On the other hand, Dolado, Motyovszki, and Pappa (2018) introduce the analysis of the impacts of monetary policy shocks on inequality by estimating an SVAR model for the US economy for the period 1979-2016, which includes the wage premium and the relative employment of skilled and unskilled workers as measures of inequality. Identification of the monetary policy shocks and estimation of the structural IRF depend on a Cholesky decomposition of the structural matrix, which implies that the Federal Funds Rate (FFR) react contemporaneously to the remaining structural shocks, and that real variables and prices react to the FFR shocks only within a month. From this initial step they find that unexpected decreases of the FFR (monetary policy expansion) increase both measures of inequality.

To rationalise the result obtained from the empirical approach, Dolado, Motyovszki, and Pappa build a NK model which, besides including search and matching frictions in the labour market, price stickiness, and heterogeneity as in Gorne-

mann, Kuester, and Nakajima (2016), includes a mechanism of asymmetry as part of the matching functions, where high-skilled workers have lower employment separation rates, and introduces capital-skill complementarity. IRF from the numerical solution of the model are similar to its empirical counterpart, and the increase in inequality generated by the expansionary monetary policy shock is explained by the new mechanisms included, since a decrease in the nominal interest rate reduces the cost of capital and, through the capital-skilled complementarity and the asymmetry of the matching function, stimulates the demand of high-skilled workers, increasing both the wage premium and the relative employment ratio.

Although there seems to be a consensus in the literature on the possible channels through which monetary policy can have effects on the distribution of wealth and income, the different approaches summarised here do not offer a unique answer on to what to expect regarding the impacts of monetary policy shocks on inequality. The answer seems to be highly conditional on the degree of heterogeneity in the sources of income; the initial conditions of inequality inside the economy; the functioning of the labour market and its specific characteristics, and on whether monetary policy includes UMP measures or not. On the other hand, apart from Villarreal (2016) and some lights given by Furceri, Loungani, and Zdzienicka (2018), the analysis seems to be highly biased towards developed economies. A little step to close this gap will be taken here by focusing on an emerging market economy like Colombia.

## 3. A Bayesian SVAR model for a small open economy

### 3.1. The reduced-form VAR model

The reduced form of the small open economy SVAR model for the colombian economy is given by:

$$\mathbf{y}_t = \boldsymbol{\alpha} + \sum_{i=1}^{\ell} \breve{}_i \mathbf{y}_{t-i} + \delta p_t^{oil} + \varepsilon_t \tag{1}$$

$$\varepsilon_t = \Gamma \boldsymbol{\nu}_t \tag{2}$$

where $\mathbf{y}_t := [y_t \ \pi_t \ r_t \ u_t \ e_t \ \rho_t]'$ is the vector of endogenous variables comprised of

the log of the real Colombian GDP ($y$); the annualised CPI inflation rate ($\pi$); the policy intervention rate of the Colombian central bank ($r$); the log of the average nominal exchange rate between the Colombian peso and the US dollar ($e$), and a measure of inequality ($\rho$). Although previous literature on the impact of monetary policy shocks in Colombia has also included a liquidity measure along with the interest rates (e.g. Gómez-Cardona 2014), I decided not to include such measure on grounds of redundancy. Given the fact that the measure of the interest rate included in the model is the instrument of monetary policy itself, any other subtle effects of liquidity on the GDP or the nominal variables should be transmitted through the interest rate equation.

Since the literature has identified unemployment as one of the main transmission channels of inequality in Colombia (Joumard and Londoño-Vélez 2013), the unemployment rate ($u$) is also included within the vector of endogenous variables. Along this five macroeconomic variables, eight different measures of inequality are separately included as the variable $\rho$. For two different measures of individual income, namely the total income per capita of the households ($ypc$), and the labour income ($labinc$), the *Gini* coefficient, the ratio of the 90th and 10th percentiles of income ($r9010$), and the ratio of the 75th and the 25th percentiles of income ($r7525$) were computed using household survey data from the Great Integrated Survey of Households (GISH) covering the period 2008-2017. Additionally, following the insights provided by Dolado, Motyovszki, and Pappa (2018) a measure of the wage premium ($premium^w$), defined as the ratio of the average values of the wages of high-skilled and low-skilled workers, was also computed in order to determine if monetary policy shocks have an impact by increasing or decreasing the burden of the business cycle stance on the unskilled labour. Under a similar line of reasoning, following the insights provided by Attanasio, Goldberg, and Pavcnik (2004) and Joumard and Londoño-Vélez (2013) for the case of Colombia, the last measure of inequality computed from household survey data is a measure of the formality premium ($premium^{for}$), defined as the ratio between the average wage of workers in the formal and informal sectors of the Colombian economy.

For the case of small open emerging economies, it has been usual in the literature to also include variables measuring the role of the terms of trade (Aguiar and Gopinath 2007; García-Rico, Pancrazi, and Uribe 2010). Uribe and Schmitt-Grohé (2017) found that, in average, the terms of trade might have explained less than 10% of the GDP fluctuations in such emerging economies, but near 20% of the

average fluctuation of the Colombian GDP. However, along the period covering the availability of the data, the dynamics of the Colombian terms of trade have been explained mostly by the fluctuations in the international price of oil and, because of being an intermediate producer of that commodity, the Colombian economy has no market power to induce changes in such price. Based on these facts, the log of the US dollar price per gallon of oil ($p^{oil}$) was included as an exogenous variable in equation (1).

All variables are available on a quarterly basis. Both the real GDP and the unemployment rate series are seasonally adjusted. For the computation of the measures of inequality, survey population weights were applied as defined by the sampling design of the GISH. Regarding the parameters of the model, $\boldsymbol{\alpha}$ is a $6 \times 1$ vector of constants; $\boldsymbol{\Phi}$ is a $6 \times 6$ matrix of reduced-form coefficents; $\boldsymbol{\delta}$ is a $6 \times 1$ vector of reduced-form coefficients associated with the exogenous variable, and $\boldsymbol{\varepsilon}_t$ is the $6 \times 1$ reduced-form vector of stochastic perturbances with covariance matrix $E\left(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'\right) = \boldsymbol{\Sigma}_{\varepsilon}$.

In equation (2), $\boldsymbol{\Gamma} \equiv \boldsymbol{\Gamma}_0^{-1}$ is a $6 \times 6$ matrix representing the structural matrix $\boldsymbol{\Gamma}_0$ of contemporaneous relations between the endogenous variables, and $\boldsymbol{\nu}$ is the vector of orthogonal structural stochastic shocks.[5] Given the stability of the dynamic system represented by equation (1) and the invertibility of the matrix $\boldsymbol{\Gamma}_0$ (Kilian and Lütkepohl 2017, Ch. 5), the Wold representation of the reduced-form model can be written as:

$$\mathbf{y}_t = \boldsymbol{\Phi}(L)^{-1} \boldsymbol{\delta} p_t^{oil} + \sum_{i=0}^{\infty} \boldsymbol{\Theta}_i \boldsymbol{\Gamma} \boldsymbol{\nu}_{t-i} \tag{3}$$

where $\boldsymbol{\Phi}(L) = \left(\mathbf{I}_6 - \sum_{i=1}^{\ell} \breve{}_i L^i\right)$ and the $6 \times 6$ matrices $\boldsymbol{\Theta}_i$ are recursive functions of the $\breve{}_i$ matrices. From equation (3), the $h$-step ahead vector of impacts of the structural shocks $\boldsymbol{\nu}_t$, or Impulse Response Functions (IRF), are determined by:

$$\nabla \mathbf{y}_{t+h}\big|_{\Delta p_{t+h}^{oil}=0} = \boldsymbol{\Theta}_h \boldsymbol{\Gamma} \tag{4}$$

[5]The structural form of the model summarised by equations (1) and (2) is $\boldsymbol{\Gamma}_0 \mathbf{y}_t = \mathbf{c} + \sum_{i=1}^{\ell} \boldsymbol{\Gamma}_i \mathbf{y}_{t-i} + \gamma p_t^{oil} + \boldsymbol{\nu}_t$. Here, $\boldsymbol{\Gamma}_0$ is a matrix containing the coefficients associated with the contemporaneous linear relationships between each one of the endogenous variables within the vector $\mathbf{y}_t$. From this structural representation, the reduced-form representation is given by $\boldsymbol{\alpha} = \boldsymbol{\Gamma}\mathbf{c}$; $\boldsymbol{\Phi}_i = \boldsymbol{\Gamma}\grave{}_i$, and $\boldsymbol{\delta} = \boldsymbol{\Gamma},$.

The relative importance of the structural shocks on the forecast error of the endogenous variables, is measured by the Forecast Error Variance Decomposition (FEVD), which depends on the $h$-step ahead matrix of Mean Squared Prediction Errors (MSPE) $\sum_{i=0}^{h-1} \mathbf{\Theta}_i \mathbf{\Gamma} \mathbf{\Sigma_\nu} \mathbf{\Gamma}' \mathbf{\Theta}_i'$, where $\mathbf{\Sigma_\nu}$ is the covariance matrix of the structural shocks. Denoting by $\eta_{k,j}^h$ the $(k, j)$ element of the $h$-step ahead MSPE matrix, the FEVD states that the contribution of the structural shock $\nu_j^*$ to the variance of the forecast error of the endogenous variable $k$, is given by the ratio:

$$\varphi_{k,j}^h = \frac{\eta_{k,j}^h}{\sum_{j=1}^{6} \eta_{k,j}^h} \tag{5}$$

Both the IRF and the contributions given by the FEVD, depend on the identification of the structural matrix $\mathbf{\Gamma}$ and the covariance matrix $\mathbf{\Sigma_\nu}$. Once $\mathbf{\Sigma_\varepsilon}$ is estimated from the reduced form in equation (1), using equation (2), it is possible to state that

$$\mathbf{\Sigma_\varepsilon} = \mathbf{\Gamma} \mathbf{\Sigma_\nu} \mathbf{\Gamma}' \tag{6}$$

The structural matrix $\mathbf{\Gamma}$ contains 36 elements and the diagonal structural covariance matrix $\mathbf{\Sigma_\nu}$ contains 6 elements. However, the estimated symmetric reduced-form covariance matrix $\mathbf{\Sigma_\varepsilon}$ only gives information about 21 elements. This overidentified system needs additional restrictions to completely identify the structural matrices and the corresponding structural shocks $\boldsymbol{\nu}$.

### 3.2. The sign restrictions approach for the identification of the structural shocks

The signs restrictions approach (Uhlig 2005) is a set identification approach of the structural matrix $\mathbf{\Gamma}$ and the IRF in equation (4). Although the main purpose is to isolate the impacts of the monetary policy (MP) shocks ($\nu^{MP}$), I follow Mountford (2005) and identify two more structural shocks, namely an aggregate demand (AD) shock ($\nu^{AD}$) and an aggregate supply (AS) shock ($\nu^{AS}$). Since the sign identifications approach is a set-identification procedure, the identification of the AD and AS structural shocks seeks to make sure that the set of possible IRF of the endogenous

variables to the MP shock is the *correct* one, based on theoretical consistency.

Leaving temporarily aside the effect of the exogenous variable $p_t^{oil}$, and assuming that $\boldsymbol{\nu}_t \geq 0$ and $\boldsymbol{\nu}_{t+h-i} = \mathbf{0}_{(6\times1)}$ for all $i \neq h$, the $h$-step ahead version of equation (3) can be rewritten as:

$$\begin{bmatrix} y_{t+h} \\ \pi_{t+h} \\ r_{t+h} \\ u_{t+h} \\ e_{t+h} \\ \rho_{t+h} \end{bmatrix} = \underbrace{\boldsymbol{\Theta}_h \boldsymbol{\Gamma}}_{\text{IRF}} \begin{bmatrix} \nu_t^{AD} \\ \nu_t^{AS} \\ \nu_t^{MP} \\ \\ \boldsymbol{\nu}^*_{t(3\times1)} \end{bmatrix} \tag{7}$$

The set identification process relies on imposing signs restrictions on the acceptable values of the matrix of IRF $\boldsymbol{\Theta}_h \boldsymbol{\Gamma}$. The restrictions imposed on the IRF of the SVAR model of the small open economy of Colombia are:

$$sign\left(\boldsymbol{\Theta}_h \boldsymbol{\Gamma}\right) = \begin{bmatrix} - & - & - & \circledast & \circledast & \circledast \\ - & + & - & \circledast & \circledast & \circledast \\ - & \circledast & + & \circledast & \circledast & \circledast \\ \circledast & \circledast & \circledast & \circledast & \circledast & \circledast \\ \circledast & \circledast & - & \circledast & \circledast & \circledast \\ \circledast & \circledast & \circledast & \circledast & \circledast & \circledast \end{bmatrix} \tag{8}$$

The third column of the matrix in equation (8) identifies the IRF of a contractionary MP shock. Theoretical and empirical consensus states that an unexpected increase in the policy intervention rate of the central bank reduces both the real GDP and the rate of inflation and generates an immediate appreciation of the nominal exchange rate. In the baseline SVAR model, the signs of the IRF of $y$ and $\pi$ to the MP shocks are maintained over a year or 4 quarters ahead, while the restriction on the sign of the response of $e$ is only imposed in the immediate or shock period. The implications of this initial assumption will be tested later by performing a robustness check.

The first and second columns of the matrix, impose sign restrictions to set-

identify the AD and the AS shocks. From a negative AD shock it is expected that both the IRFs of the real GDP and inflation have negative signs, and an immediate reduction in the intervention policy rate as a possible systematic monetary policy response to the shock. On the other hand, a negative AS shock or *cost-push* inflation shock is expected to generate a negative response of the real GDP and an increase in the inflation rate. In the particular case of the AS shock, a systematic MP response is not expected, since the monetary authorities should not, in principle, respond to this kind of short run *cost-push* shocks. The $\circledast$ symbols in the matrix, indicate that there are no restrictions on the sign of the corresponding IRFs. Since the objective is to determine the impact of monetary policy shocks on inequality, the variables related to it are let free to respond given the three identified structural shocks, including the MP shock.

### 3.3. The Bayesian method

Defining $\boldsymbol{\Psi} := [\boldsymbol{\alpha}\ vec(\boldsymbol{\Phi}_1) \ldots vec(\boldsymbol{\Phi}_\ell)\ \ \boldsymbol{\delta}]'_{6(\ell+2)\times 1}$; and the set $data = \{\mathbf{y}_0, \ldots \mathbf{y}_T, p_0^{oil}, \ldots, p_T^{oil}\}$ from the reduced-form VAR model in equation (1), the idea of the Bayesian approach applied to the SVAR of the small open economy, is to first obtain the *posterior* distribution of the parameters $\boldsymbol{\Psi}$ and $\boldsymbol{\Sigma}_\varepsilon$ $-g(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_\varepsilon|data)-$from the process of updating the *prior* distribution of such parameters $-g(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_\varepsilon)-$using the likelihood function of the data $-\mathscr{L}(data|\boldsymbol{\Psi}, \boldsymbol{\Sigma}_\varepsilon)-$and applying the Bayes rule. Such updating process is described by the proportional equation:

$$g(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_\varepsilon|data) \propto \mathscr{L}(data|\boldsymbol{\Psi}, \boldsymbol{\Sigma}_\varepsilon)\, g(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_\varepsilon) \tag{9}$$

For the case of small samples, McNeish (2016) recommends to use informative priors in order to overcome the small-sample issues of frequentist methods relying on asymptotics. In the case of VAR models, one of the most used informative priors, and the one chosen to generate the posterior distribution for the small open economy VAR model of the Colombian economy, is the independent normal-Wishart prior, since it takes both $\boldsymbol{\Psi}$ and $\boldsymbol{\Sigma}_\varepsilon$ as unknowns, with no assumed dependence between the variance of the residuals $\varepsilon$ and the coefficients $\phi$. Under this assumptions, the process of updating the prior through equation (9) does not have an analytical solution, and depends on a numerical method called the *Gibbs* sampling method: After $M$ iterations of random draws from the conditional

distributions $g(\boldsymbol{\Psi}|\boldsymbol{\Sigma}_{\varepsilon}^{(0)}), g(\boldsymbol{\Sigma}_{\varepsilon}|\boldsymbol{\Psi}^{(1)}), \ldots, g(\boldsymbol{\Psi}|\boldsymbol{\Sigma}_{\varepsilon}^{(M-1)}), g(\boldsymbol{\Sigma}_{\varepsilon}|\boldsymbol{\Psi}^{(M)})$, a sufficient $K$ number of iterations is burned, and it is expected for the remaining $M-K$ conditional distribution draws to converge to draws from the unconditional distributions $g(\boldsymbol{\Psi})$ and $g(\boldsymbol{\Sigma}_{\varepsilon})$. Independence of $\boldsymbol{\Psi}$ and $\boldsymbol{\Sigma}_{\varepsilon}$ implies that $g(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_{\varepsilon}) = g(\boldsymbol{\Psi})g(\boldsymbol{\Sigma}_{\varepsilon})$. It is usual to define $M = 20000$ and $K = 10000$.[6]

The Monte Carlo sampling nature of the Bayesian method, makes it a perfect framework to carry on the set-identification procedure given by the sign restrictions approach. After the posterior has been determined for the reduced-form VAR model parameters, from the draws of this posterior distribution, it is possible to generate the posterior distribution of the IRF defined by equations (7) and (8) (Arias, Rubio-Ramírez, and Waggoner 2018). From the identification equation (6), assuming unity variance of the structural shocks ($\boldsymbol{\Sigma}_{\boldsymbol{\nu}} = \mathbf{I}_6$), the orthogonal reduced-form parameterisation is defined as

$$\boldsymbol{\Sigma}_{\varepsilon} = h(\boldsymbol{\Sigma}_{\varepsilon})\, h(\boldsymbol{\Sigma}_{\varepsilon})' \tag{10}$$

$$\boldsymbol{\Gamma} = h(\boldsymbol{\Sigma}_{\varepsilon})\, \mathbf{Q} \tag{11}$$

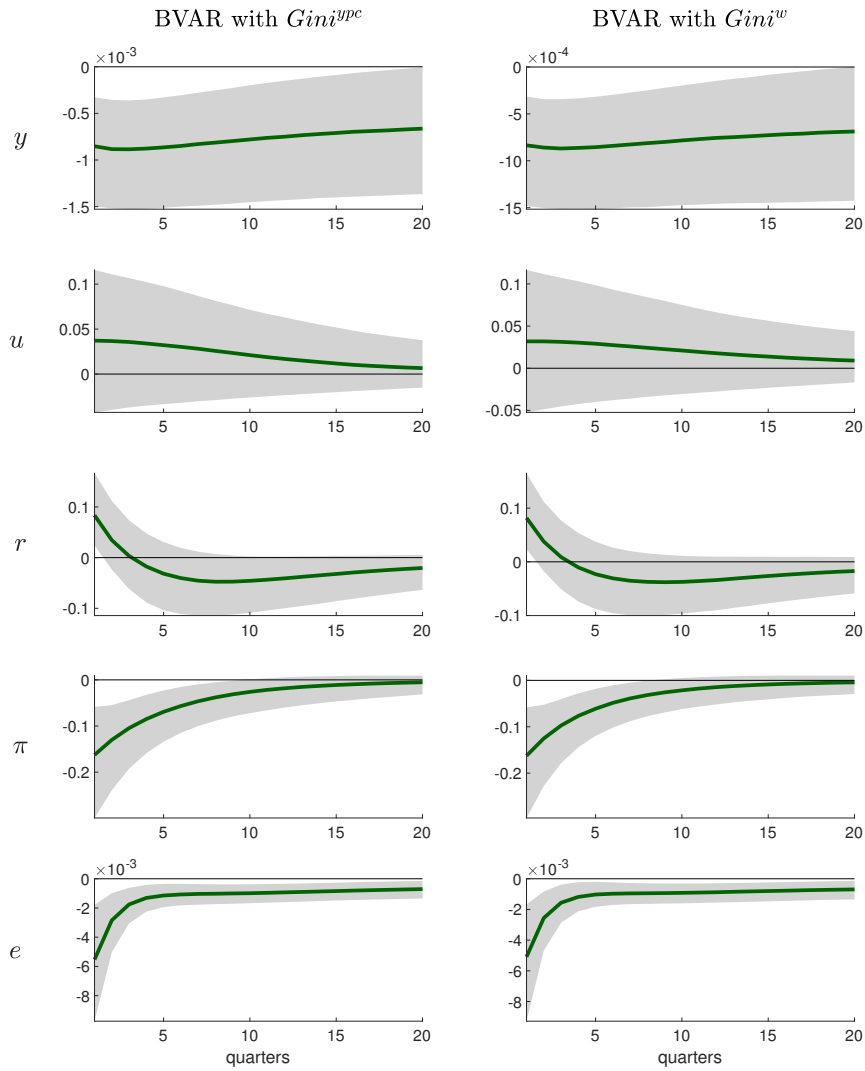
with $\mathbf{Q}$ an orthogonal matrix such that $\mathbf{Q}\mathbf{Q}' = \mathbf{I}_6$ and where the natural candidate for $h(\boldsymbol{\Sigma}_{\varepsilon})$ is the low triangular matrix $\mathbf{P}$ that comes from the Cholesky decomposition of $\boldsymbol{\Sigma}_{\varepsilon}$. Based on equations (10) and (11), algorithm 1 in Arias, Rubio-Ramírez, and Waggoner (2018) applied to the small-open economy SVAR model for Colombia is:

a. Draw $(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_{\varepsilon})$ from $g(\boldsymbol{\Psi}, \boldsymbol{\Sigma}_{\varepsilon}|data)$
b. Draw a matrix $\mathbf{Z}_{6\times 6}$ with each element coming from an standard normal distribution and perform the QR decomposition $\mathbf{Z} = \mathbf{QR}$, with the diagonal of $\mathbf{R}$ normalised to be positive.
c. Keep $(\boldsymbol{\Psi}, \boldsymbol{\Gamma})$ if the restrictions in $sign(\boldsymbol{\Theta}_h \boldsymbol{\Gamma})$ (equation (8)) are satisfied.
d. Return to step 1 until the required number of draws in the Gibbs sampler are obtained.

The median of the draws $\{\boldsymbol{\Theta}_{h,j}\boldsymbol{\Gamma}_j\}_{j=K+1}^{M}$ are the punctual estimators of IRF. The credible bands are defined by the 16th and 84th percentiles of the posterior distribution (68% credible sets).

[6] See Kadiyala and Karlsson (1997) for a detailed explanation of the method.

Figure 1. IRF of the macroeconomic variables to an MP shock




## 4. On the impacts of monetary policy shocks

Figure 1 depicts the IRF for the six endogenous macroeconomic variables included in the vector **y** of endogenous variables of the Bayesian SVAR(1) models with inequality measures $Gini^{ypc}$ and $Gini^{w}$. As expected from the sign restrictions imposed on the identification of the MP structural shock, after an unexpected one deviation increase of the intervention policy rate of the Colombian central bank, negative responses from the real GDP and the inflation rate are observed for a minimum of four quarters (1 year), along with an immediate appreciation of the nominal exchange.

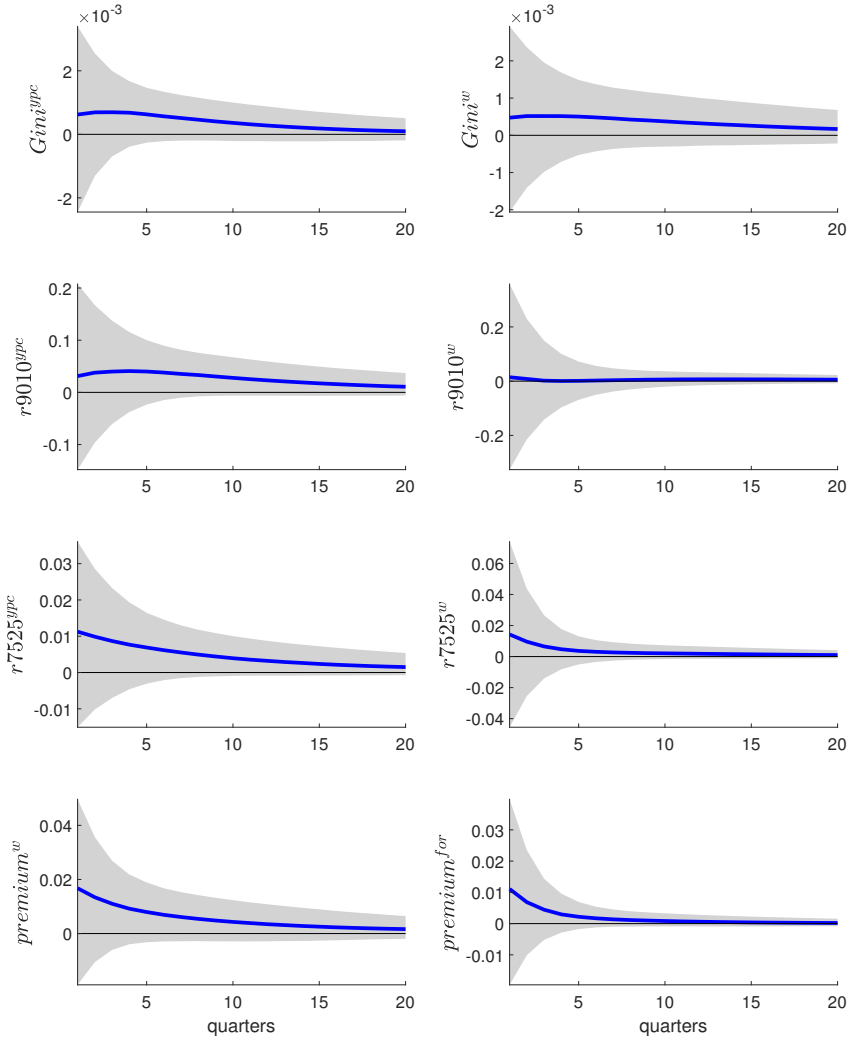
However, beyond the imposed structure on the IRF, the negative effect on the real GDP of the contractionary MP shock persists for at least 10 more quarters, whilst the policy effect on inflation remains for at least 5 more quarters. These results were replicated by the eight SVAR estimated models.[7] The contractionary effects of the MP shock are then quite persistent in the case of the Colombian economy, a result that has been documented before by the previous literature on the effects of monetary policy shocks in Colombia (Fernández 2003; Gómez-Cardona 2014). The contractionary effect seems not to be clear regarding the response of the unemployment rate. Although the direction of the median response is as expected, namely an increase after the contractionary MP shock, there is a lot of uncertainty around the true value of such response, as shown by the 68% credible bands. Another interesting effect is the persistent appreciation of the Colombian exchange rate as a result of the MP shock, in spite of the sign restriction only imposing a negative response in the same period of the shock. This is a striking result regarding the noncompliance of the *uncovered interest parity* theory, from which a depreciation of the nominal exchange rate would be expected right after an appreciation shock.[8]

Figure 2 depicts the estimated IRF of the eight different measures of inequality to a one standard deviation structural contractionary MP shock in the eight different Bayesian SVAR(1) models. In terms of direction, the contractionary MP shock generates a positive median response in almost all measures of inequality. The median effect of a unexpected contractionary increase of the intervention rate of the Colombian central bank is to increase income inequality. At least in terms of the median response, this finding seems to be in the line of that of Furceri, Loungani, and Zdzienicka (2018), for which increases in inequality out of contractionary MP shocks are most likely to occur in economies with low redistribution (high inequality). The median response seems to be relatively more important for the Gini coefficient of the total income per capita of the household ($Gini^{ypc}$); the Gini coefficient of the labour income ($Gini^{w}$); the ratios of the 90th to the 10th ($r9010^{ypc}$) and the 75th to the 25th ($r7525^{ypc}$) percentiles of total income per capita of the household, and the wage premium ($premium^{w}$). For each of these five measures, the 68% credible sets show that a positive response is more likely

[7] See Appendix B.

[8] Previous literature, also based on the use of VAR-type models, has found mixed evidence concerning the compliance of the uncovered parity theory for the case of the Colombian economy. See for instance Echavarría, López, and Misas (2008) and Gómez-Cardona (2014).

Figure 2. IRF of the inequality measures (ρ) to an MP shock




to occur 4 quarters (one year) after the shock, given a significant reduction in the uncertainty around positive values. In spite of this, zero remains to be a credible value of the true response.

It is interesting to note that the stronger median responses are associated to the measures of inequality based on total income per capita. In the case of the measures based on labour income, the stronger response is that of the wage premium, followed by the response of the Gini coefficient which is three times lower that its counterpart based on the total income per capita.

A more formal approach to determine the relative importance of the MP shocks to explain variations in the inequality measures, is the FEVD summarised in

Table 1. FEVD for the different measures of income inequality in each of the estimated Bayesian SVAR(1) models

| ρ | $h = 5$ | | | | $h = 10$ | | | | $h = 20$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\nu^{MP}$ | $\nu^{AD}$ | $\nu^{AS}$ | $\boldsymbol{\nu}^*$ | $\nu^{MP}$ | $\nu^{AD}$ | $\nu^{AS}$ | $\boldsymbol{\nu}^*$ | $\nu^{MP}$ | $\nu^{AD}$ | $\nu^{AS}$ | $\boldsymbol{\nu}^*$ |
| $Gini^{ypc}$ | 15 | 19 | 15 | 51 | 15 | 18 | 15 | 52 | 15 | 17 | 16 | 52 |
| $Gini^{w}$ | 15 | 17 | 16 | 52 | 16 | 17 | 16 | 51 | 16 | 16 | 17 | 52 |
| $r9010^{ypc}$ | 15 | 16 | 16 | 53 | 16 | 16 | 16 | 51 | 16 | 16 | 17 | 52 |
| $r9010^{w}$ | 14 | 15 | 17 | 54 | 14 | 16 | 17 | 53 | 14 | 16 | 17 | 53 |
| $r7525^{ypc}$ | 17 | 16 | 16 | 51 | 18 | 16 | 16 | 51 | 17 | 15 | 17 | 51 |
| $r7525^{w}$ | 15 | 18 | 16 | 52 | 15 | 18 | 16 | 51 | 15 | 18 | 16 | 51 |
| $premium^{w}$ | 18 | 17 | 15 | 50 | 18 | 17 | 15 | 50 | 17 | 16 | 16 | 51 |
| $premium^{for}$ | 15 | 17 | 16 | 52 | 15 | 17 | 16 | 51 | 15 | 17 | 16 | 51 |

equation (5). Table 1 presents the median proportion of the variance of the forecast error of these measures explained by each of the structural shocks identified by the sign restrictions imposed by equation (8) in the $h$-step ahead quarter after the shock. At least in average, the MP shock ($\nu^{MP}$) explains the lesser proportion –less than 20% –of the variance of the forecast error of the inequality variables . The exceptions are the ratio of the 75th to the 25th percentiles of the total income per capita of the household, and the wage premium, for which the MP shock explain relatively more of the forecast error variance than the aggregate demand shock ($\nu^{AD}$) or the aggregate supply shock ($\nu^{AS}$). Within the set of inequality measures for which a response to the contractionary MP shock is more likely to be positive, such response seems to be relatively more important for the variables $r9010^{ypc}$ and $r7525^{w}$, a result that might imply that monetary policy tends to increase the income differences between the households at the extremes of the income distribution. Nevertheless, since the unidentified shocks explain most of the variance, the effect of monetary policy shocks on inequality is indeed quite small.

### 4.1. A peek on the possible impacts of systematic monetary policy

Technically speaking the IRF and FEVD analysis permits only to establish the response of the endogenous variables to random structural shocks. In the case of monetary policy, the shock is related to unexpected or surprise changes in the intervention policy rate of the central bank. However, the sign identification of the aggregate demand structural shock offers a possible way to take a look on the possible impacts of systematic monetary policy on inequality. From the IRF of the

endogenous variables to the AD shock ($\nu^{AD}$) in Figure 3, it is possible to see the nature of the implicit reaction function of the Colombian central bank implied by the SVAR(1) models.

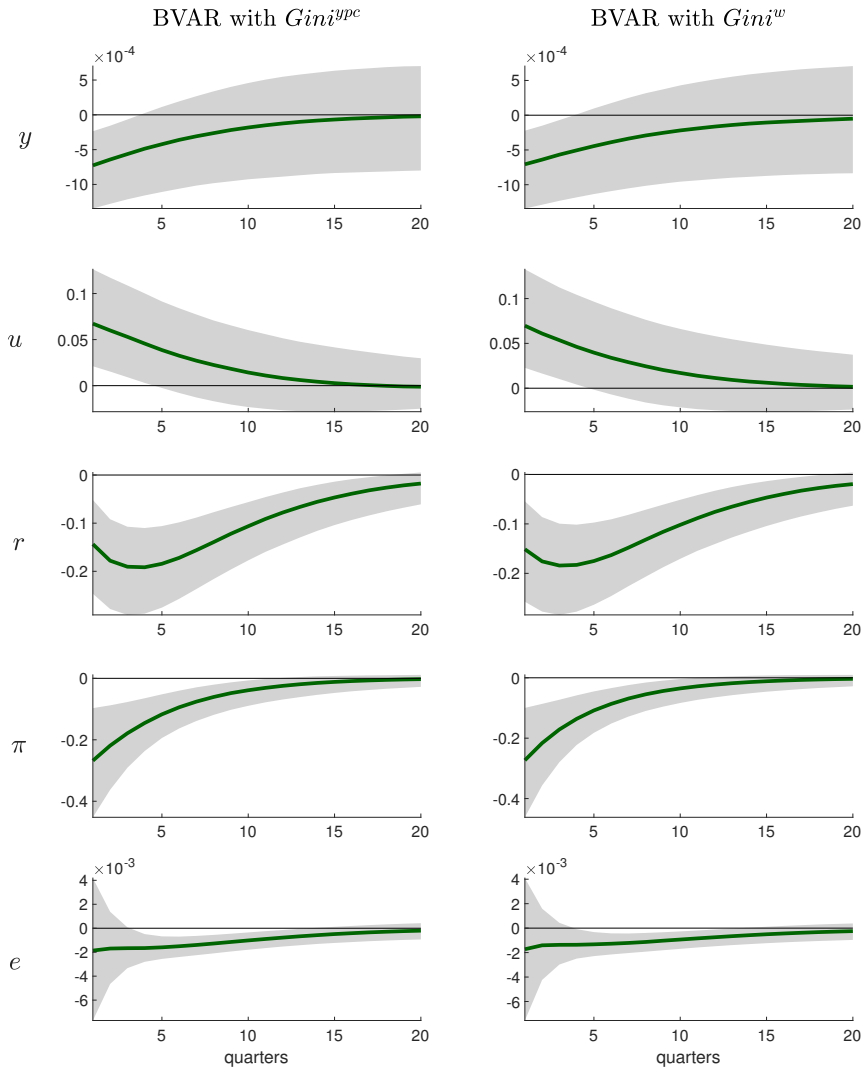
Only in the quarter of the shock, the sign restriction imposes a negative response of the intervention policy rate plus an immediate reduction in the inflation rate. The IRF of the intervention rate for both models presented in the figure show that whilst real GDP recovers from the shock in the first 5 quarters, the intervention rate level slowly increases while inflation reaches its steady state level 10 quarters ahead. This kind of slower response of inflation had been already identified in the case of the monetary policy shock. The estimated implicit reaction function of the central bank depicted by the IRF, represents the nature of the countercyclical or systematic monetary policy that the monetary authority in Colombia has been trying to implement since the adoption of the inflation targeting regime in the early 2000s.[9]

Although it is not possible to associate the endogenous response of the inequality measures to the pure systematic component of monetary policy, the IRF shown by Figure 4 involve it for sure, and through them, we could have been taking a peek on the contribution of the countercyclical stance of the central bank of Colombia on the dynamics of inequality. In the period after the shock, when the implicit reaction function of the central bank operates by slowing the pace of the reduction of its policy rate, a positive response is observed after the fifth quarter for the wage premium, the ratio of the 75th to the 25th percentiles of the labour income, and the premium of formality. A similar but much more uncertain response is observed in the rest of inequality measures. It is then possible for contractionary systematic monetary policy to have a negative effect on inequality.

The sign restrictions approach framed by the Bayesian method has provided us with weak evidence regarding the distributional consequences of monetary policy shocks in the case of a highly unequal emerging economy like Colombia. Although the sign of the median response of inequality measures point to the direction of contractionary MP shocks having a negative impact on income inequality, a significant amount of uncertainty surrounds the true value of such response, which might be zero as well. At the same time, after the identification of the implicit action of the central bank reaction function once an aggregate demand

[9]This pattern of responses was resembled by the IRF of the eight SVAR(1) models estimated. See Appendix C.

Figure 3. IRF of the macroeconomic variables to an aggregate demand shock

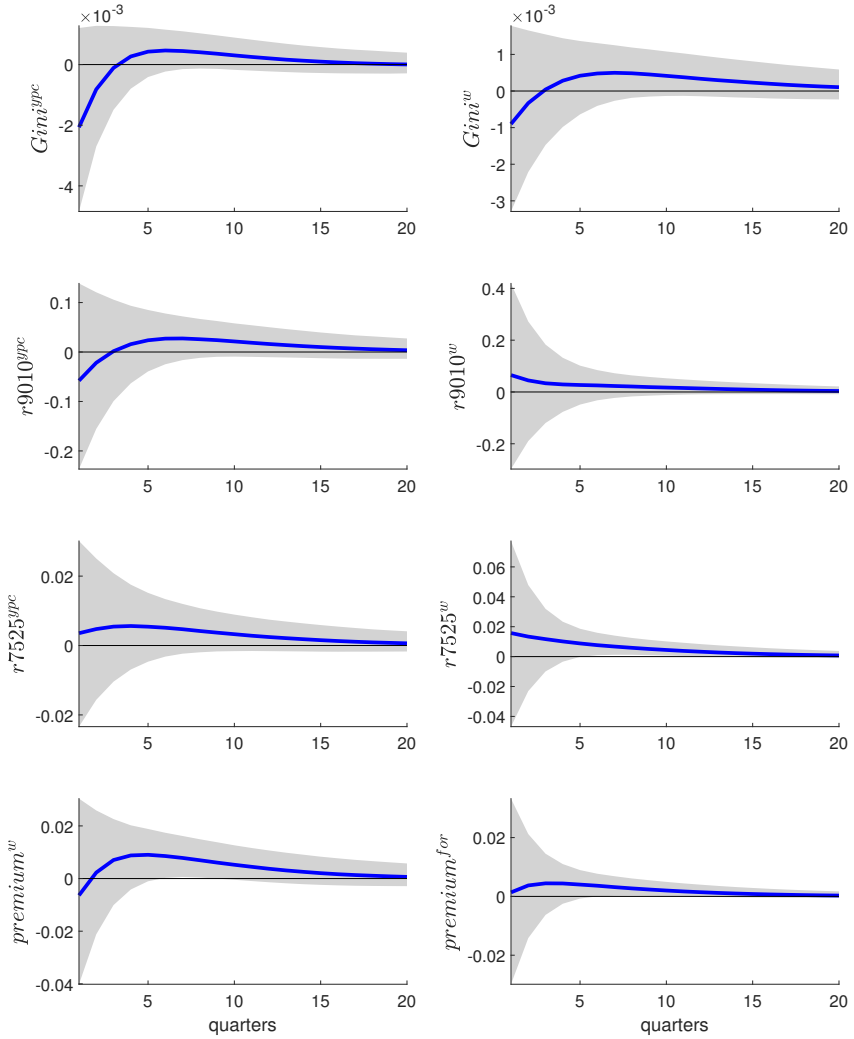



shock have been realised, it is also likely for a contractionary systematic response through the intervention policy rate to have negative impacts on inequality. In spite of the weak evidence, though, it is worth to carry out some robustness checks on the results, either to give support or to put some more doubt on the results of the baseline model.

## 5. Robustness checks

Two different robustness checks were carried out. The first one is associated with the number of quarters ahead of the impact period of the MP shock, for which

Figure 4. IRF of the inequality measures (ρ) to an aggregate demand shock




the sign restrictions on the IRF of real GDP and inflation are set to be kept. In the baseline model such restrictions are kept for four quarters, namely in the impact period ($h$ = 0), and 3 more quarters ahead $h$ = 3. It is worth to check if the number of $h$ periods for which such restrictions are imposed, affect somehow the main results already obtained. A second and more structural check is related to the way that the structural orthogonal shocks are identified, in particular the monetary policy shock, and how this affects the results of the baseline model. Instead of the sign restrictions approach, we re-estimate the posteriors of the reduced-form models and the IRF by following a more restrictive approach called triangular factorisation. Under this approach, zero restrictions are imposed only

on the immediate impacts of the structural shocks.

### 5.1. Periods of the sign restrictions

Figure 5, depicts the IRF of the $Gini^{ypc}$ and $Gini^{w}$ measures of inequality to a contractionary MP shock, for the cases in which the sign restrictions on the IRF of both real GDP and the inflation rate are kept only for the impact period ($h = 0$); for the impact period and one quarter ahead ($h = 1$), and for the impact period and two quarters ahead ($h = 2$). As shown in the first row of the figure, for the case of the measures of inequality based on total income per capita, the number of periods for which the sign restrictions of the MP shocks persist, does not seem to have a significant influence on their median IRF. However, to those inequality variables based on the labour income, the median response is heavily affected.[10] Although after the 5th quarter after the shock the median impact of the contractionary MP shock on inequality is negative, between the impact period and the 4th quarter ahead of the shock this median impact is, on the contrary, negative. The pattern clearly changes once the sign restrictions are kept for one year, as can be seen in Figure 2.

Notwithstanding this robustness issue in the median response of the inequality variables based on labour income, the uncertainty around the true value of the response being positive is high in the first quarters after the shock for all measures of inequality, as it was also in the baseline model. This result in particular is robust.

### 5.2. An alternative way to identify the monetary policy shocks

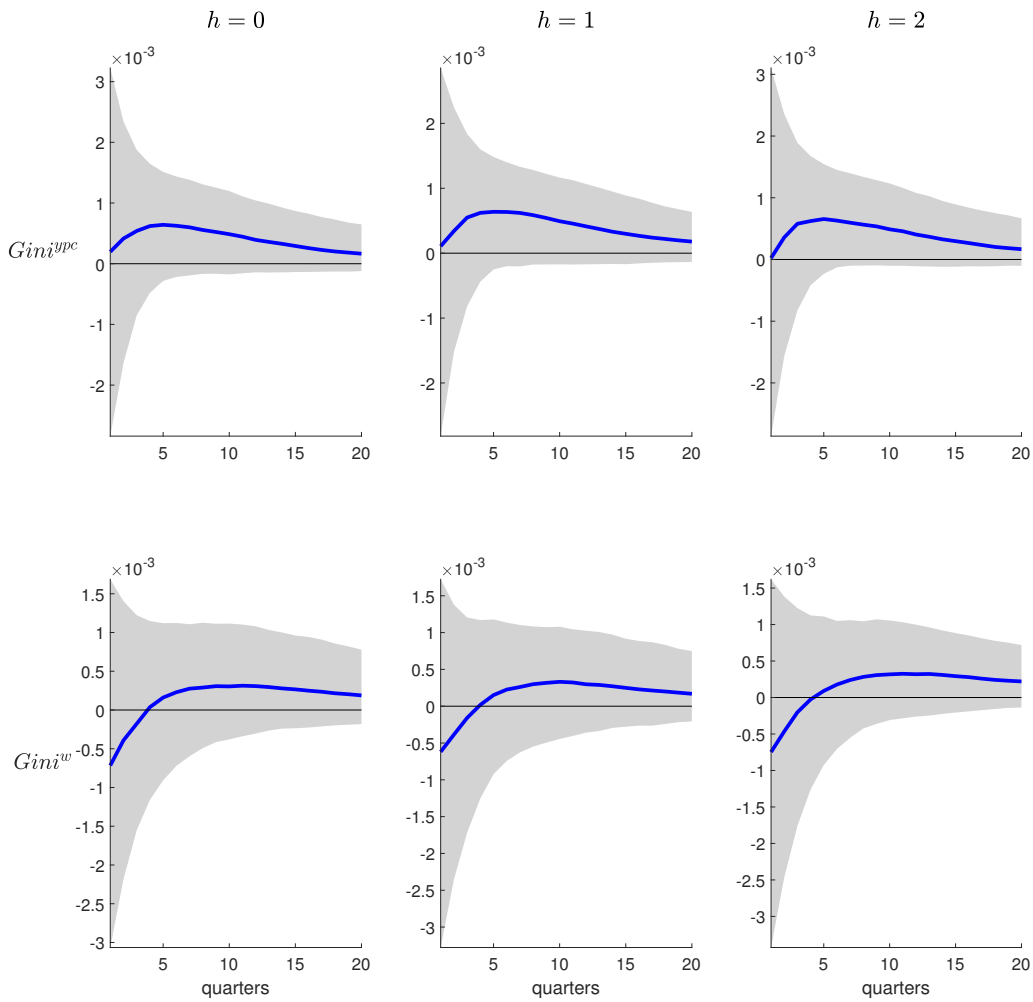
Taking as a initial step the Cholesky factorisation of the covariance matrix $\mathbf{\Sigma}_{\varepsilon} = \mathbf{P}\mathbf{P}'$, the triangular factorisation approach (Hamilton 1994), identifies both $\mathbf{\Gamma}$ and $\mathbf{\Sigma}_{\nu}$ in equation (6), using the fact that

$$\mathbf{\Gamma}\mathbf{\Sigma}_{\nu}^{1/2} = \mathbf{P} \tag{12}$$

This alternative way of identification of the structural shocks depends on the order of the variables in the vector of endogenous variables in the reduced-form equation (1) of the SVAR model. To be as close as possible to the theoretical impli-

[10] See also Appendix D.

Figure 5. IRF of the inequality measures (ρ) to a contractionary MP shock with sign restrictions on $y$ and $\pi$ kept for $h$-quarters ahead




cations implied by the sign restrictions approach, we re-define $\mathbf{y} := [y\ \pi\ r\ e\ u\ \rho]'$. The identified structural matrix $\mathbf{\Gamma}$ is a low-triangular matrix that imposes zero restrictions on the immediate effects of the 6 implicit structural shocks, such that:

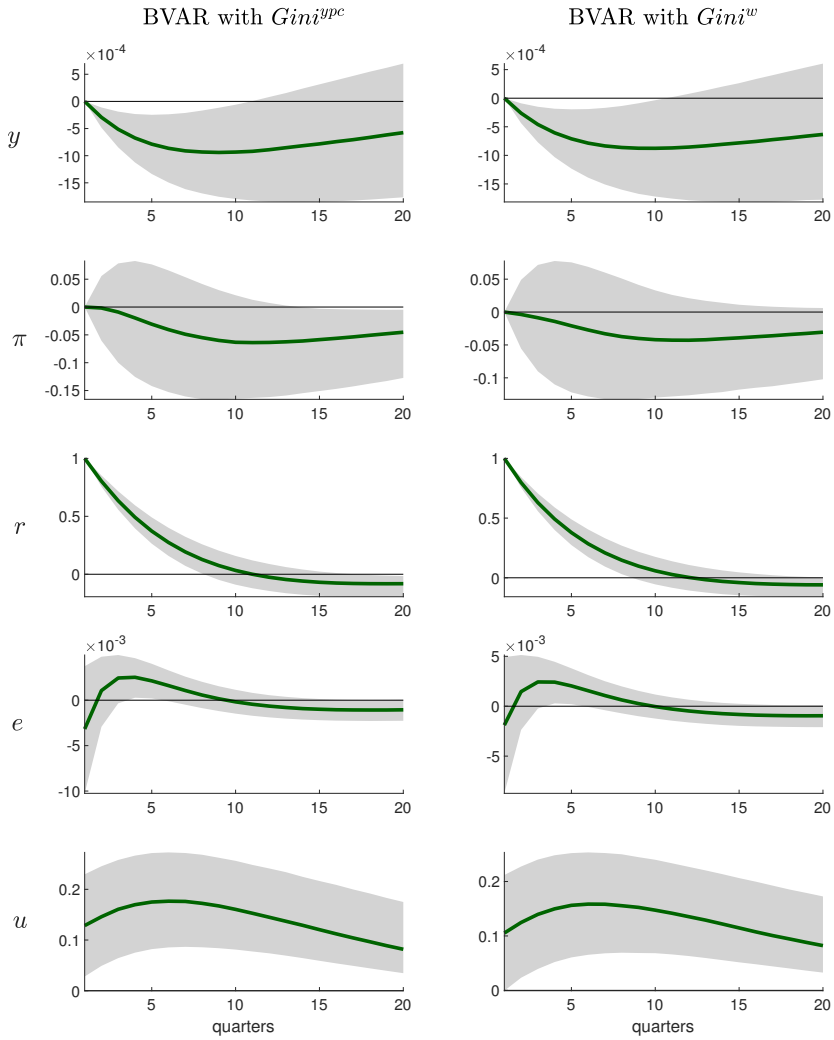
$$
\begin{bmatrix} y_{t+h} \\ \pi_{t+h} \\ r_{t+h} \\ e_{t+h} \\ u_{t+h} \\ \rho_{t+h} \end{bmatrix} = \mathbf{\Theta}_h \begin{bmatrix} \Gamma_{11} & 0 & 0 & 0 & 0 & 0 \\ \Gamma_{21} & \Gamma_{22} & 0 & 0 & 0 & 0 \\ \Gamma_{31} & \Gamma_{32} & \Gamma_{33} & 0 & 0 & 0 \\ \Gamma_{41} & \Gamma_{42} & \Gamma_{43} & \Gamma_{44} & 0 & 0 \\ \Gamma_{51} & \Gamma_{52} & \Gamma_{53} & \Gamma_{54} & \Gamma_{55} & 0 \\ \Gamma_{61} & \Gamma_{62} & \Gamma_{63} & \Gamma_{64} & \Gamma_{65} & \Gamma_{66} \end{bmatrix} \begin{bmatrix} \nu_t^y \\ \nu_t^\pi \\ \nu_t^r \\ \nu_t^e \\ \nu_t^u \\ \nu_t^\rho \end{bmatrix} \tag{13}
$$

The order of the variables in vector $y$ and the low-triangular form of the identified structural matrix, implies that in the period of the shock ($h = 0$), the MP shock, identified in the third column of $\mathbf{\Gamma}$, does not have an immediate impact in both the GDP and the inflation rate, but might have a non-zero impact on the rest of endogenous variables. Some *hard to defend* relationships are implied here, as the one that implies that a shock in the unemployment rate does not have an immediate impact on real GDP or inflation. It is the presence of this questionable assumptions what makes the sign restrictions approach more flexible. Nevertheless, it is important to contrast the robustness of the results presented under this alternative approach, especially those regarding the IRF of the different inequality measures to the contractionary MP shock.

The triangular factorisation is also performed under the Bayesian framework. For each draw $\mathbf{\Psi}$ and $\mathbf{\Sigma}_\varepsilon$ from the posterior of the reduced-form model in equation (1), a Cholesky decomposition of the latter is followed by the triangular factorisation in equation (12). Using the values of the draws of $\mathbf{\Psi}$, the draws of the IRF are obtained once the matrix $\mathbf{\Gamma}$ has been identified. The median draw of the IRF is the point-wise estimator of the true response of the endogenous variables to the structural shocks, and the 68% credible sets/bands are defined by the corresponding percentiles of the IRF draws. It must be clear that the magnitudes of the IRF using the two identification methods are not comparable, since in the case of the sign restrictions the shock is equivalent to a one standard deviation increase in the intervention policy rate, whilst in the case of the triangular factorisation, the procedure implies a one unit (one percent) change in the policy instrument of the Colombian central bank.

The estimated median IRF of the five endogenous macroeconomic variables and their 68% associated credible sets are depicted in Figure 6. The IRF patterns

Figure 6. IRF of the macroeconomic variables to a MP shock ($\nu^r$) under the triangular factorisation identification approach




obtained were similar for all the eight models estimated, and the figure presents only the results for the models with $Gini^{ypc}$ and $Gini^{w}$ representing the endogenous measure of inequality $\rho$.[11] At first sight it is clear that important differences emerge once the identification procedure is changed. The contractionary effect of the MP shock on the real GDP seems to be less persistent than in the sign restrictions case. On the other hand, the median response of the inflation rate is negative, but the credible sets show that it is more likely for the impact to be negative twelve quarters (three years) after the shock. A response quite different to that presented

[11]See Appendix E.

by the estimated IRF under the sign restrictions.

Another important difference is related to the response of the unemployment rate. Under the triangular factorisation approach, the contractionary MP shock has a strong, negative and long-run persistent impact on unemployment, in contrast to the uncertain effect observed under the sign restrictions approach. Perhaps the most striking difference emerges from the median IRF of the nominal exchange rate. Identification of the MP shock using the triangular factorisation implies a median response consistent with the compliance of the uncovered interest parity theory, since on impact an appreciation is generated, and it is immediately followed by a depreciation that persists for at least two years. Nevertheless, credible sets only support the depreciation of the exchange rate from the fourth quarter after the shock.

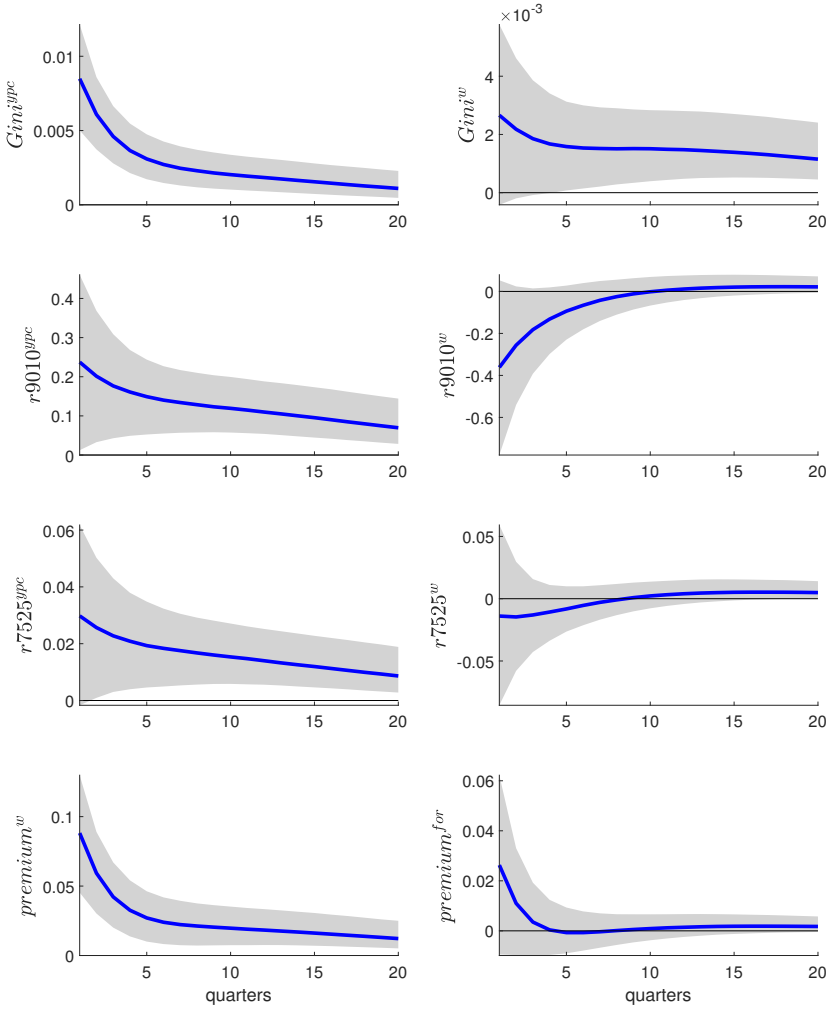
Table 2. Proportion of variance of the FE of ρ explained by the MP shock under the triangular factorisation identification approach

| ρ | $h = 5$ | $h = 10$ | $h = 20$ |
|---|---|---|---|
| $Gini^{ypc}$ | 13 | 13 | 13 |
| $Gini^{w}$ | 3 | 4 | 5 |
| $r9010^{ypc}$ | 4 | 5 | 6 |
| $r9010^{w}$ | 2 | 3 | 3 |
| $r7525^{ypc}$ | 3 | 4 | 5 |
| $r7525^{w}$ | 1 | 2 | 2 |
| $premium^{w}$ | 9 | 9 | 10 |
| $premium^{for}$ | 2 | 2 | 2 |

Figure 7 depicts the estimated median IRF and credible sets of the measures of inequality to the MP shock. For five of these variables, namely $Gini^{ypc}$; $Gini^{w}$; $r9010^{ypc}$; $r7525^{ypc}$ and $premium^{w}$, the median IRF shows that a contractionary MP shock increases inequality. In contrast with the results presented under the sign restrictions approach, the credible 68% sets indicate that the true value of the response is positive and quite persistent along a period of at least 20 quarters. However, the result is not robust to all the measures. It is possible that the increase of inequality after the contractionary MP shock is related to the positive and persistent negative effect on unemployment under this identification procedure.

Regarding the relative importance of the MP shock to explain the forecast error variance of inequality, Table 2 shows the explained proportions for each one of

Figure 7. IRF of the inequality measures (ρ) to a contractionary MP shock under the triangular factorisation identification approach




the ρ variables. Only for the case of the Gini coefficient of the total income per capita of the households, the proportion is higher than 10% and its magnitude is kept for 20 quarters ahead. For the premium wage, the proportion barely reaches 10% after 5 years, whilst for the rest of inequality measures the value is not greater than 6%. These values are quite smaller than their counterparts computed under the sign restrictions approach.

# 6. Conclusions

Following the Bayesian approach, SVAR models are estimated for the case of Colombia, a small open emerging economy with high inequality. The sign restrictions methodology for the identification of structural shocks defines the impulse response function (IRF) analysis that seeks to determine the impact of monetary policy (MP) shocks on inequality. Using household survey data for the period between the first quarter of 2008 and the fourth quarter of 2017, eight different measures of inequality were computed based on two different individual income measures, and each one of these was included separately in the vector of endogenous variables of the SVAR model.

The sign restrictions imposed on the IRF of the baseline model, identify three structural shocks, namely a MP shock, an aggregate demand shock and an aggregate supply shock. For a small open economy like Colombia, regardless of the inequality measure included in the vector of endogenous variables of the SVAR model, the draws of the posterior distribution of the IRF show a persistent contractionary effect of an unexpected increase in the intervention policy rate of the Colombian central bank. However, the evidence is weak in terms of the true value of the response of the unemployment rate to the contractionary MP shock, in spite of the median response being positive as expected. Regarding the response of the exchange rate, the IRF analysis shows no support to the uncovered interest parity theory.

In this baseline model, the median IRF of all inequality measures show that after a contractionary MP shock inequality tends to increase, supporting a result on the literature stating that for economies with low redistribution, this kind of shocks should indeed make worse the inequality burden. However, once the credible sets are taken into account, the evidence on the true direction of the response significantly weakens, making the response more likely to be positive two years after the shock. Although it might true that the impact of a contractionary MP shock on inequality is negative, the relative importance of the monetary policy shocks to explain the future variance of the forecast errors of the measures of inequality is quite small according to the FEVD analysis.

From the identification of the structural aggregate demand shock, it was possible to establish the existence of an implicit reaction function of the Colombian central bank and, through it, to make a conjecture about the possible effect of a sys-

tematic contractionary stance of the monetary authority. It is quite possible in the case of the Colombian economy that even a systematic monetary contractionary response has a medium-term negative effect on inequality.

The results of the baseline model are based on two key factors. First, the number of quarters ahead of the MP shock for which the sign restrictions are kept for the IRF of the real GDP and the rate of inflation and, second, the particular structural identification approach selected. Two robustness contrasts were performed to control for these factors. In the first case, by contrasting the baseline results with estimations imposing the restrictions only in the quarter of the shock; in the quarter of the shock and one quarter ahead of the shock, and in the quarter of the shock and two quarters ahead of the shock, the results in terms of the IRF of inequality to the contractionary MP shock seem to be robust for the measures based on the total income per capita, but not for those based on the labour income.

Regarding the structural identification approach, an alternative procedure called triangular factorisation was performed under the Bayesian framework. This procedure identifies a low-triangular structural matrix implying zero restrictions only on the immediate impacts of the structural shocks on a set of endogenous variables, which are going to be defined by the order of the endogenous variables inside the vector containing them. A particular order was selected in order to get a close approximation to the sign restrictions imposed in the baseline model. There are important differences between the two approaches. First, the IRF analysis using the triangular factorisation, shows that the contractionary effect of the MP shock is less persistent, and shows a really weak response of the inflation rate. Second, with respect to the sign restrictions approach, the negative effect on unemployment is quite significant and highly persistent using the contrasting method. Although the result is not robust for all the inequality measures, the triangular factorisation makes a strong case for the support of a negative effect of the contractionary MP shock on inequality. This last result may be highly related to the significative and negative response of the unemployment rate.

The evidence presented is not strongly conclusive about the effects of the monetary policy shocks on inequality in Colombia. Nevertheless, it shows that even if the effect is negative, it is quite possible that its magnitude and relative importance on the dynamics of inequality is quite small. It makes sense that in emerging open economies like Colombia, where inequality has been so persistent and high for a long period of time, other structural factors aside macroeconomic policy are

much more important. However, it is also possible that the small sample issue, that was at least partially taken care of the use of a Bayesian framework, keeps on being important, or that beyond a simple empirical approach, it is necessary to include a strong theoretical basis to define the workings of the possible different transmission channels of monetary policy on the distribution of wealth and income in emerging economies.

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## Appendix A. Optimal lag selection

The Deviance Information Criterion (DIC) is a Bayesian information criterion for model selection Spiegelhalter et al. (2002). Following the notation from the reduced-form VAR model in (1), and the Bayesian update procedure of the prior distribution in (9), the DIC criterion is computed as:

$$DIC = E_{\boldsymbol{\Psi},\boldsymbol{\Sigma}_\varepsilon|data}\left[-2\ln\mathscr{L}\left(data|\boldsymbol{\Psi},\boldsymbol{\Sigma}_\varepsilon\right)\right] + p_D$$

The first term is called the Bayesian Deviance, and it is a measure of fit, whilst the second term, $p_D$, is a measure of model complexity, and its purpose is to look for the effective number of parameters, and depends on the posterior mean of the deviance.

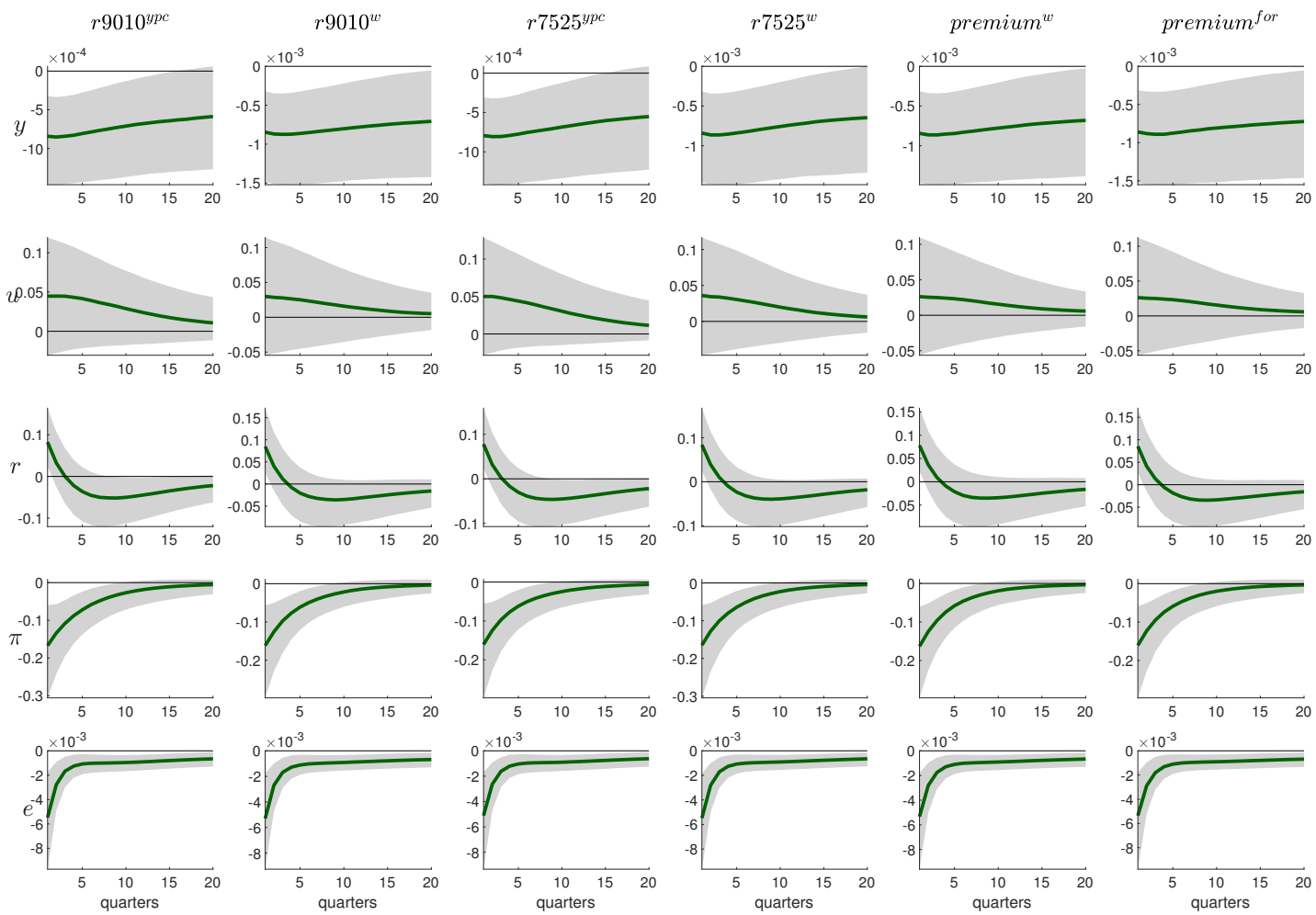
Table A.1. DIC values for the reduced-form VAR($\ell$) models with $Gini^{ypc}$

| Reduced-form model | DIC |
|---|---|
| VAR (8) | 4897.77 |
| VAR (7) | 4745.86 |
| VAR (6) | 4898.07 |
| VAR (5) | 5401.86 |
| VAR (4) | 1220.81 |
| VAR (3) | -90.64 |
| VAR (2) | -474.86 |
| VAR (1) | -644.73 |

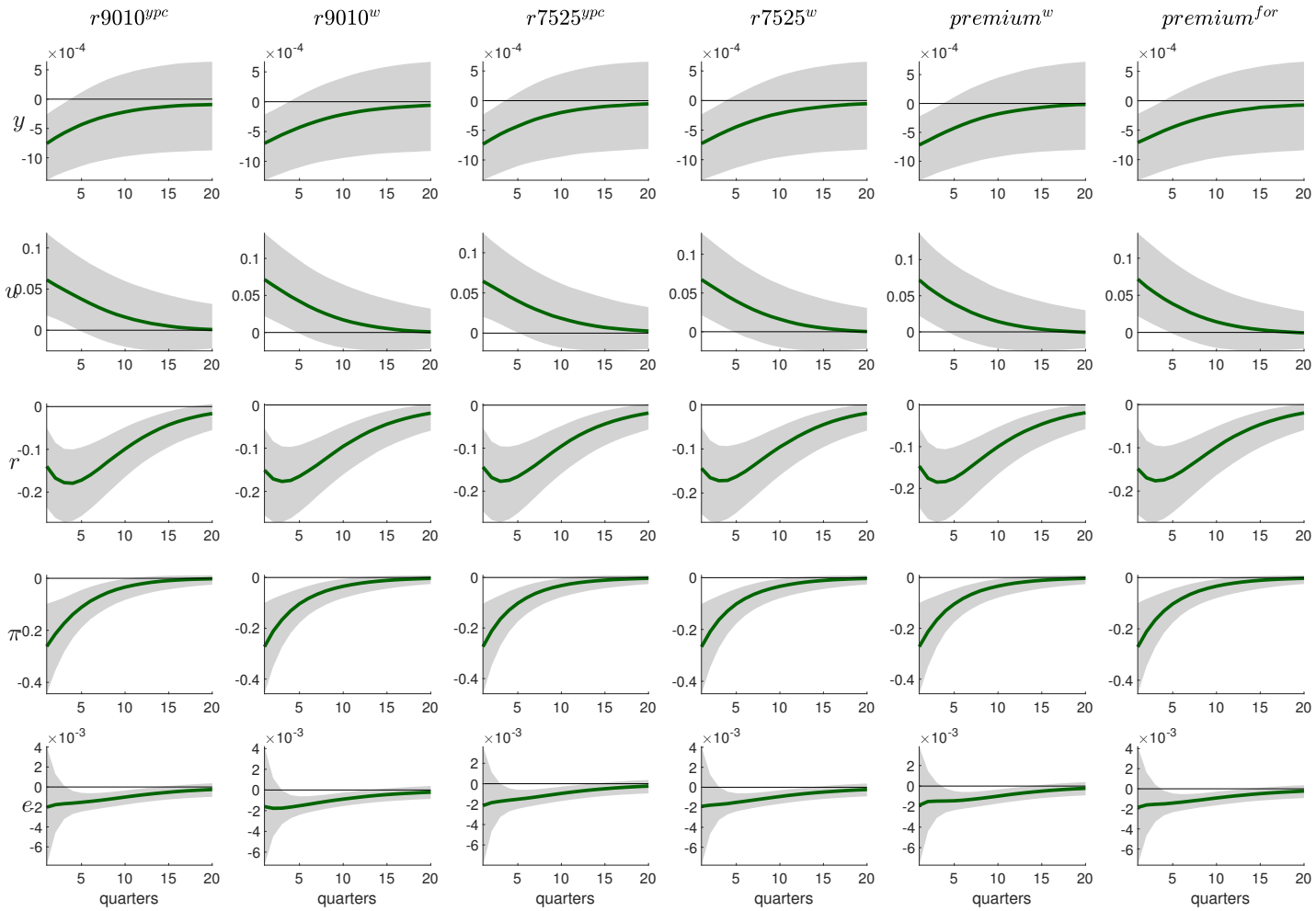
Notice that the better the model fits the data, the larger the likelihood, and the smaller the value of the DIC criterion. From the DIC values referenced in Table A.1, for the different reduced-form VAR($\ell$) specifications for the SVAR model for

the Colombian economy with measure of inequality $\rho := Gini^{ypc}$, it is possible to conclude that $\ell^* = 1$ minimises the DIC criterion.

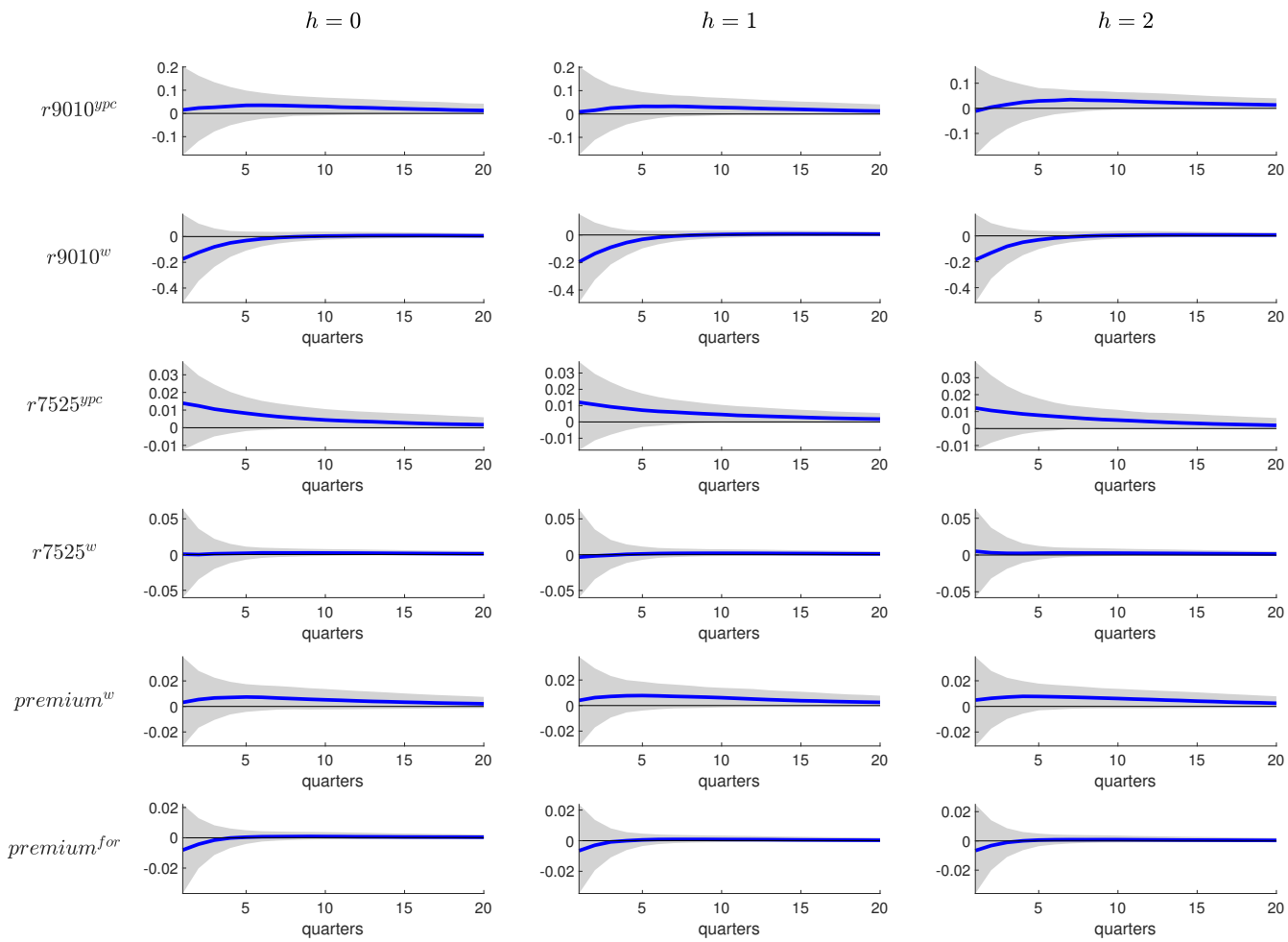
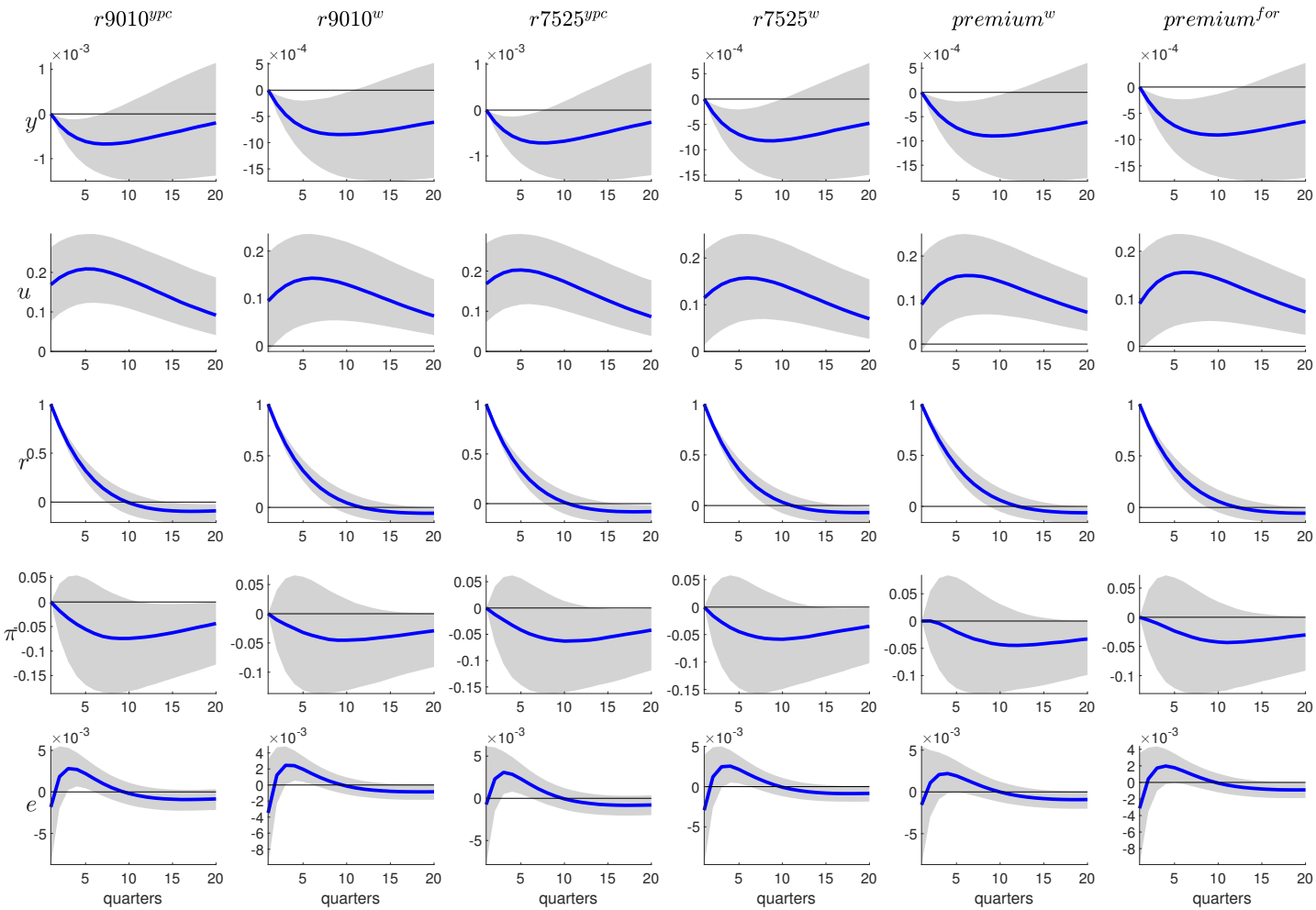
# Appendix B. Sign restrictions: IRF of macro variables to a MP shock



# Appendix C. Sign restrictions: IRF of macro variables to an AD shock



# Appendix D. Robustness: $h$-quarters IRF of measures of inequality to a MP shock



# Appendix E. Triangular factorisation: IRF of macro variables to a MP shock